



IOT PROTOCOL



AI ENABLED

AI SMART HELMET

Prevent Accidents. Protect Workers.

AI-powered real-time hazard detection system for industrial safety

● Array Active



⚡ 0ms Latency



Telemetry: Syncing

Swipe to explore the system 🖱️



The Silent Threats in Heavy Industry

Every year, thousands of workplace accidents result from unnoticed environmental hazards and sudden equipment collisions.

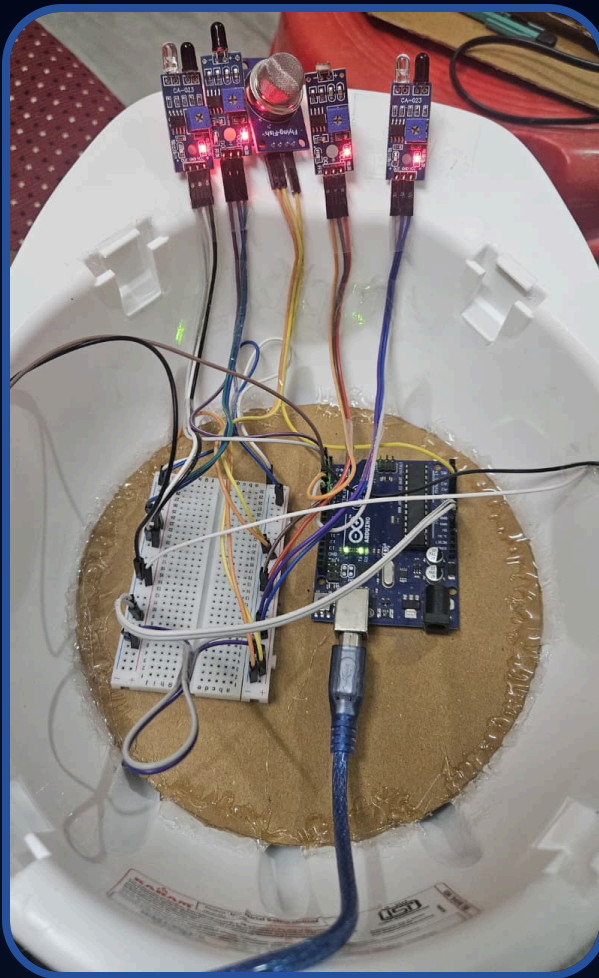
Traditional safety gear is purely passive—high-risk zones demand **proactive intelligence.**



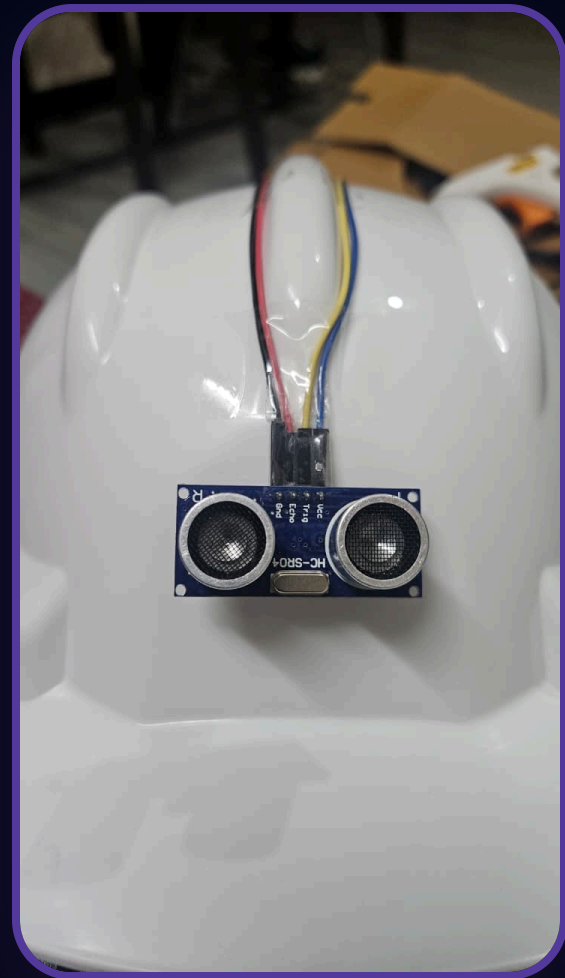
Active Hazard Intelligence



Internal Embedded Assembly



External Sensor Layout



A zero-latency smart helmet that actively maps surroundings and alerts workers instantly before danger strikes.

4-Layer Embedded Protection Architecture

Driven by highly-optimized C/C++ core logic.



Toxic Gas

MQ-2 Sensor for early fume warnings.



Thermal Target

IR sensors & LM35 to prevent heat stress.

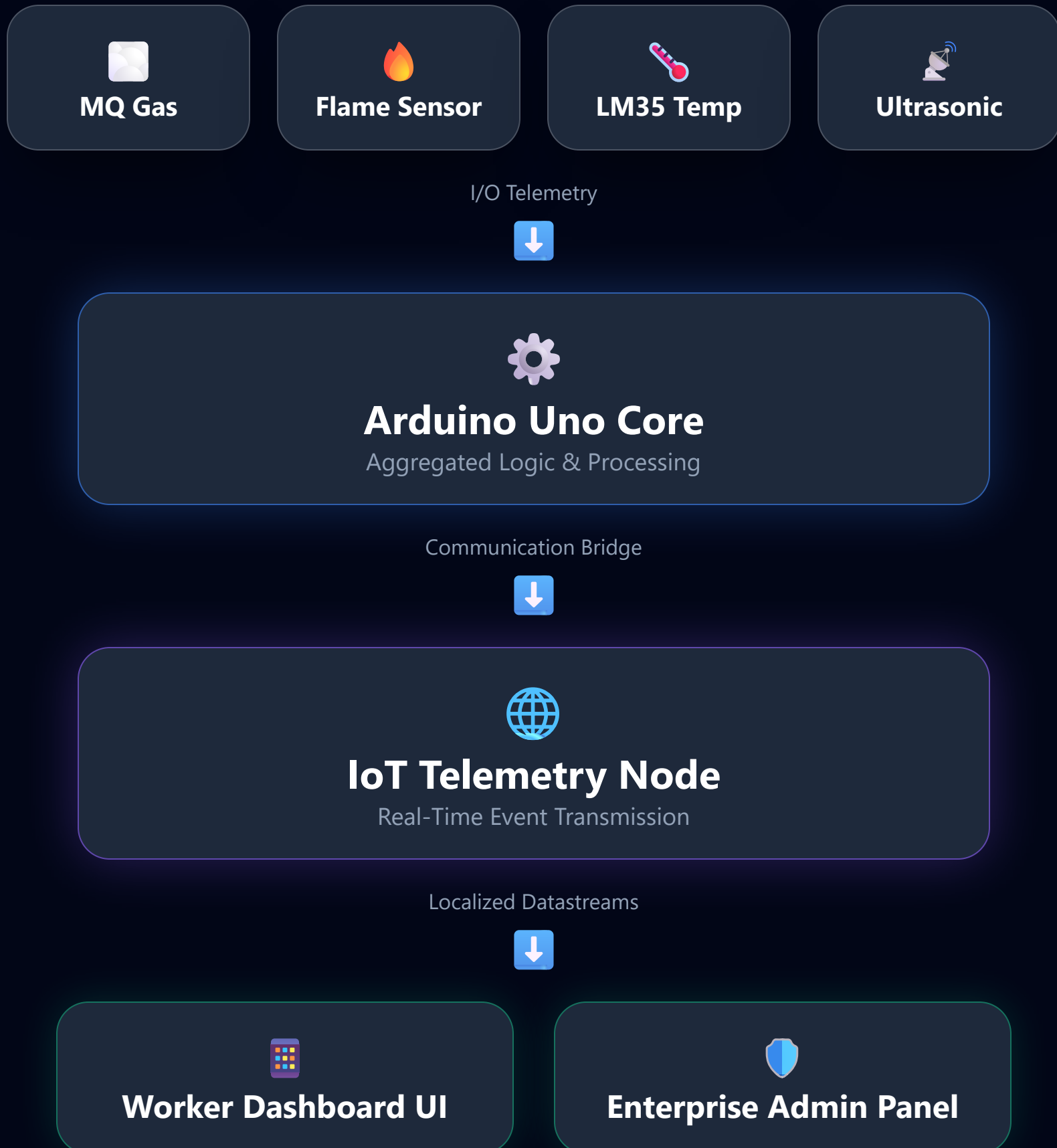


Proximity Defense Array

HC-SR04 Ultrasonic spacing arrays.

System Architecture

Real-time data flow from sensors to the dashboard.

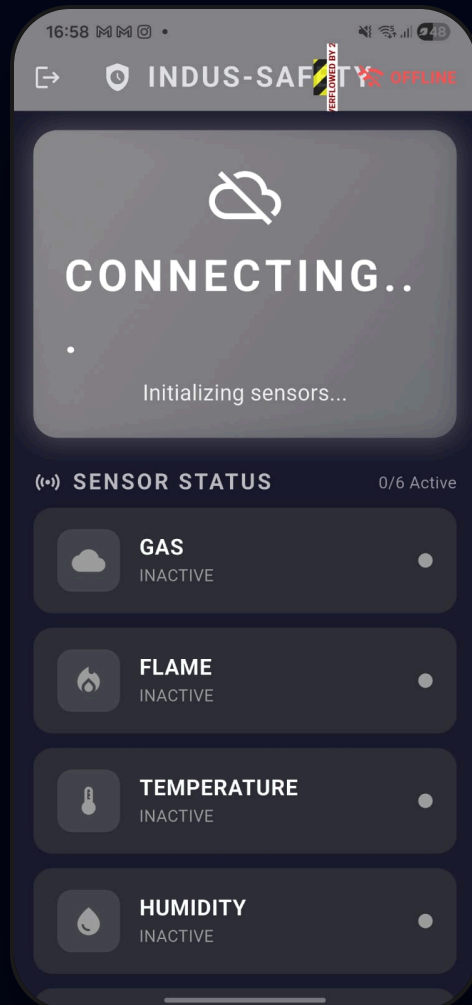


Real-Time Actionable Telemetry

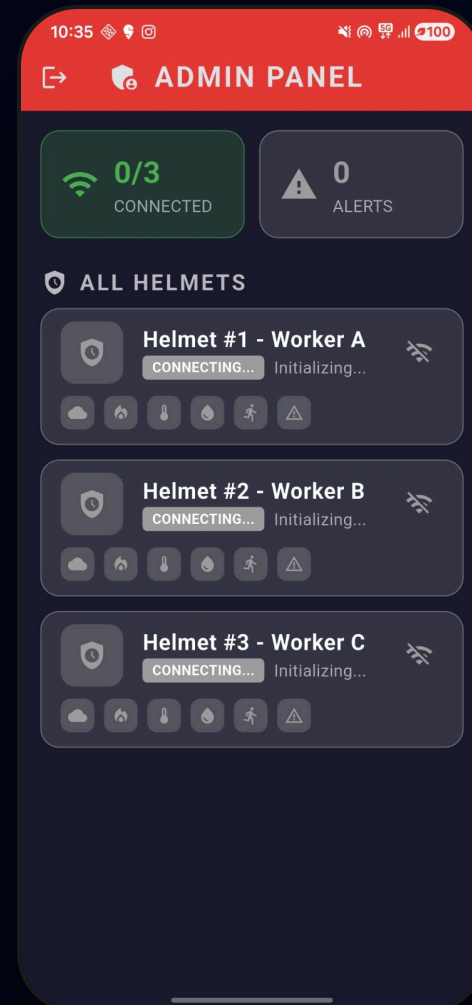
A high-performance **Flutter** app providing instant, real-time visual dashboards.



Worker Monitoring



Admin Dashboard





Built to protect lives using **AI + IoT**



C/C++



Arduino Uno



Flutter



Dart



Embedded Systems



Hardware Sensors



Serial Comms



GitHub repository link in
comments!